- **Advantages:**
    - Simplifies complex problems into smaller, manageable sub-problems.
    - Provides clear, concise code for problems like traversing hierarchical structures.
- **Disadvantages:**
    - Can lead to stack overflow if the base case is not defined or if there are too many recursive calls.
    - Generally less efficient due to function call overhead and repeated calculations.

Optimizing Recursion:

Memoization: Caching results of expensive function calls to avoid redundant computations.

```
def fibonacci_memo(n, memo={}):
    if n in memo:
        return memo[n]
    if n <= 1:
        return n
    memo[n] = fibonacci_memo(n - 1, memo) + fibonacci_memo(n - 2, memo)
    return memo[n]

print(fibonacci_memo(50))
```